

Week 6 Lecture Plan — Complex Survey Analysis: Why NHANES Is Not a Simple Random Sample

Duration: 2.5 h = 150 min (Tue, before the Thu lab).

Timing: the Slide 1–16 markers below sum to **147** — 142 min of content plus the 5-minute break at Slide 9 — leaving **3 min of float** in the session. The deck (`slides/Week6_survey_slides.qmd`) spends that float on its opening Continuity recap, which this plan does not separately budget. **Slide 15** (pitfalls checklist, 6 min) recaps material already covered and is the designated cut if the mechanics slides 4–8 run long.

Learning objectives

- Explain why NHANES is a stratified, multistage, clustered probability sample — not a simple random sample — and why that breaks default standard errors.
- Identify the three design objects every NHANES analysis needs: weights (WTINT / WTMEC / WTSAF), strata (SDMVSTRA), and PSU (SDMVPSU).
- Choose and pool the correct weight when combining survey cycles, and state what design effects do to confidence intervals.
- Contrast the MASLD paper’s unweighted estimates with design-aware ones, and articulate why unweighted was a *choice*.
- Reason about the fasting-vs-non-fasting weight puzzle that MASLD’s own analyte mix creates.

Continuity

- Recap Wk5: the targeted IV-vs-III $\times$ sex test gave a ratio of HRs of 1.00 (0.42–2.35) — underpowered rather than resolved. Today: the **weight follows the measurement subsample** (CDC’s least-common-denominator rule), and **domain analysis** preserves the parent design — build the design, then `subset()`.
- Today: a different threat entirely — not confounding or modification, but whose *population* the numbers describe. The paper analyzed NHANES as if it were a simple random sample.
- Sets up L5 (naive vs survey-weighted estimation) and P4, your first design-aware re-estimate of your own paper.

Slide 1 — Where we are in the project (5 min)

- Timeline recap: Replicate $\rightarrow$ Interrogate $\rightarrow$ Improve $\rightarrow$ Defend. M1 is Wk8; **M2 (Wk9) is your critique of another group’s M1**; M3 (Wk11) is your individual defense.
- P1 eligibility, P2 Table 1, P3 effect modification are done — all built on an *unweighted* analytic file.
- Today we question a design assumption baked into every number so far.
- MASLD hook: the paper reports HRs to two decimals but does not name the survey design.

Slide 2 — The motivating puzzle (8 min)

- Show the paper’s Group IV all-cause HR ~ 15.1 (unadjusted). Ask: 15.1 *for whom* — the sample, or US adults with MASLD?
- The gap between “sample” and “population” is exactly what survey design bridges.
- Preview the punchline: design-aware weighting attenuates Group IV toward ~ 12.4 and widens the CI.
- MASLD hook: the paper’s inference target is implicitly the US population, but its math is the sample’s.

Slide 3 — What “simple random sample” assumes (10 min)

- SRS: every person independently, equally likely to be drawn; default SEs assume this.
- Two violations coming: unequal selection probabilities (weights) and non-independence (clustering).
- Consequence of ignoring them: point estimates biased toward the sample composition; SEs usually too *small*.

- MASLD hook: default `coxph` SEs in the replication silently assume SRS.

Slide 4 — How NHANES is actually built (12 min)

- Multistage: (1) counties (PSUs), (2) segments, (3) households, (4) persons — selection at each stage.
- Stratified first by design domain; oversampling of targeted subgroups (e.g., older adults, some race/ethnicity groups, low income).
- Result: neither equal probabilities nor independent draws — the two SRS assumptions both fail by design.
- MASLD hook: MASLD adults are not spread randomly across PSUs; they cluster with the households sampled.

Slide 5 — Weights: WTINT, WTMEC, WTSAF (12 min)

- A weight $\sim$ how many people in the target population this respondent represents (inverse selection probability, adjusted for nonresponse and post-stratification).
- Three tiers by how far into the exam a variable lives: WTINT (interview) $\supseteq$ WTMEC (mobile exam) $\supseteq$ WTSAF (fasting subsample).
- Rule: use the weight for the *most restrictive* data source your analysis actually requires.
- MASLD hook: FLI needs triglycerides (a fasting analyte) $\rightarrow$ the honest weight is WTSAF, the smallest, most-adjusted one.

Slide 6 — Strata and PSUs: SDMVSTRA, SDMVPSU (10 min)

- SDMVSTRA encodes the design strata; SDMVPSU encodes the (masked) primary sampling units within a stratum.
- These drive *variance*, not the point estimate: clustering means neighbors resemble each other, so effective sample size $< N$.
- Analytic strata/PSU are pseudo-values released for disclosure control — use as provided, never re-derive.
- MASLD hook: none of SDMVSTRA / SDMVPSU appear anywhere in the paper’s methods.

Slide 7 — Pooling weights across cycles (10 min)

- NHANES 2007–2018 = six 2-year cycles; the paper stacks them.
- Weights are built per cycle $\rightarrow$ you must divide by the number of cycles pooled (here /6) so the weighted total isn’t multiplied.
- Use the survey-cycle-appropriate weight variable and pool consistently; strata/PSU are pooled as-is.
- MASLD hook: with six cycles combined, the correct pooled fasting weight is WTSAF/6, a step easy to miss.

Slide 8 — The design effect (10 min)

- DEFF $\sim$ variance under the real design $\div$ variance under SRS; typically > 1 because of clustering + weighting.
- Practical read: when DEFF > 1 , your *effective* N is smaller than your row count; honest CIs get wider.
- “N = 6,300” overstates the information content once design is respected.
- MASLD hook: the paper’s tight CIs partly reflect pretending each of $\sim 6,300$ rows is an independent draw.

Slide 9 — Short break (5 min)

- Stretch. Coming up: what happens to the MASLD numbers when we do it right, and the fasting-weight puzzle.

Slide 10 — “Unweighted” is a choice, not a default (10 min)

- Some analysts argue for unweighted, model-based inference when the outcome model is richly adjusted (conditional, not descriptive, target).
- But that must be *stated and defended* — silence reads as an oversight, not a decision.
- Weighting and covariate adjustment are separate axes, not two rival framings. Design decides which population the estimate speaks for; adjustment decides which covariates it holds fixed. Every combination is available, and the honest report crosses them (Slide 11).
- Declare the estimand — population-average, conditional on covariates, or both — name the weight you used and why, and say which cell of the 2×2 your headline number came from.
- MASLD hook: the paper never declares which target it’s after, so readers can’t tell if unweighted was reasoned or accidental.

Slide 11 — What weighting does to MASLD (12 min)

- Apply WTSAF (pooled) + SDMVSTRA + SDMVPSU via a survey-aware Cox fit.
- Group IV all-cause HR moves ~15.1 -> ~12.4 unadjusted; CIs widen from the design effect.
- Direction of the headline survives — Group IV still worst — but the magnitude and precision are honestly smaller/wider.
- Then cross the axes and show all four cells, fitted on the *same* Model-1 complete-case rows (n = 5,939): read **down** a column for adjustment, **across** a row for design. Holding the rows fixed is the only thing that makes a difference attributable to the axis you moved — and it is why the crude IV-vs-I cell reads 16.05 rather than the ~15.1 of the full analytic file.
- IV vs I, the paper’s contrast: unweighted crude 16.05 (5.91–43.58), unweighted Model 1 3.79 (1.38–10.39); weighted crude 11.93 (2.60–54.84), weighted Model 1 2.97 (0.69–12.79).
- IV vs III, our locked contrast: unweighted crude 2.34 (1.72–3.18), unweighted Model 1 1.06 (0.77–1.45); weighted crude 3.07 (2.11–4.48), weighted Model 1 1.29 (0.90–1.86).
- The familiar ledger that runs unadjusted, then fully adjusted, then survey-weighted mixes crude and adjusted models fitted on different row sets. It is not a sequence of successive corrections and you do not present it as one.
- MASLD hook: on the paper’s own IV-vs-I contrast, design decides whether the interval excludes 1; on the locked IV-vs-III contrast, adjustment dominates and design moves the estimate in the opposite direction and by less. “Survey design matters” is true of one contrast and much weaker for the other.

Slide 12 — The question settled by CDC’s least-common-denominator rule: fasting vs non-fasting analytes (12 min)

- FLI combines triglycerides + GGT + BMI + waist. TG is a *fasting* analyte (WTSAF domain); GGT is *not* (available in the broader MEC, WTMEC).
- Which weight is **primary** is settled by CDC’s least-common-denominator rule, not contested: use the weight of the smallest applicable subsample.
- The rule selects WTSAF (FLI needs a fasting analyte). MEC is not a defensible *primary* — it would weight to a population in which the exposure cannot be constructed — but a MEC-weighted run, **explicitly labelled as a sensitivity analysis**, shows students how much the rule was worth. The point is to apply the rule, say so, and quantify what it bought.
- MASLD hook: MASLD’s own ascertainment recipe forces a weight decision the paper never confronts.

Slide 13 — Subset the design, never the data (10 min)

- Cardinal rule: define the full-survey design object once, then *subset within it* (e.g., `subset()` on the design), never filter rows first and re-declare a design on the survivors.
- Dropping rows first discards the strata/PSU information that non-selected units contribute to variance -> wrong SEs.
- This is the #1 practical error students will hit in L5.

- MASLD hook: MASLD eligibility ($FLI \geq 60$, ≥ 1 cardiometabolic criterion) is a *subset*, so it must be applied to the design, keeping WTSAF/strata/PSU intact.

Slide 14 — Design-aware workflow, end to end (8 min)

- Steps: (1) pick weight tier by most-restrictive variable, (2) pool /cycles, (3) declare design with weight+strata+PSU, (4) subset within design, (5) fit survey-aware model, (6) report DEFF-widened CIs.
- Report a one-line “design statement” in every table caption — the habit that would have fixed the paper.
- Sensitivity analysis is part of the deliverable, not optional polish. For the weight that means: WTSAF primary by the rule, WTMEC reported alongside it and labelled as sensitivity.
- MASLD hook: this six-step recipe is precisely what P4 asks you to run on your paper.

Slide 15 — Common pitfalls checklist (6 min)

- Forgetting to divide pooled weights by the number of cycles.
- Filtering rows before declaring the design (the fatal one).
- Using WTMEC when a fasting analyte is in the model, or vice versa.
- Reading unweighted vs weighted divergence as “error” rather than as two different estimands.

Slide 16 — Bridge to today’s lab (L5) and P4 (7 min)

- In L5 (EpiMethods Complex Survey module) you will re-fit the MASLD Group I–IV Cox model two ways on the fixed dataset: a *naïve* unweighted fit, then a *design-aware* fit declaring WTSAF/6 + SDMVSTRA + SDMVPSU, subsetting the design (not the rows) to the MASLD-eligible sample. You will tabulate naïve vs survey HRs and CIs side by side and quantify the attenuation ($\sim 15 \rightarrow \sim 12.4$) and CI widening.
- This becomes **P4, your design-aware re-estimate**: apply the same six-step workflow to *your* paper’s headline estimate, declare and justify your chosen weight (naming the fasting-vs-non-fasting tension if it applies), report the naïve-vs-weighted contrast, and write the one-line design statement your paper omitted. Those six steps are the survey route. A group whose paper qualified at M0 criterion 5 on a time-to-event structure instead has no weight to declare and therefore no naïve-vs-correct number to report: their P4 is to reproduce the paper’s survival estimand keeping follow-up time and censoring, then interrogate the one structural choice they named at M0 — the time scale, the censoring rule, the index date, or how competing events were handled — and report what that choice does to the estimate and to its interpretation. Be exact about what is comparable: weighted and unweighted fits of the same survival model on the same rows are one estimand on one scale, so the gap between them is attributable to the design; a survival model set beside a collapsed “ever died” model is two different estimands on two different scales, so the gap between them measures the collapse, not the design, and is never the design-aware correction. P4 is formative and feeds directly into your M1 replication and M2 critique of another group’s M1.